Parth Ganeriwala, Ph.D.

SOFTWARE ENGINEER II · RESEARCH SCHOLAR
Avidyne | ASSIST Research Lab, Florida Institute of Technology
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Summary

Ph.D. computer scientist researching formal verification and assurance of learning-enabled autonomous systems, with 32 peer-reviewed publications spanning transfer learning, computer vision, LLM-based tooling, and formal methods. Currently a Software Engineer II at Avidyne, building DO-178C safety-critical avionics software in embedded C++ – pairing production engineering discipline with applied ML research.

Education

Florida Institute of Technology — *Melbourne, Florida*
PH.D. IN COMPUTER SCIENCE: GPA - 4.0/4.0 — *January 2023 - May 2026*

- Research Interests: Formal Methods, Artificial Intelligence, Machine Learning, Deep and Transfer Learning, LLMs and Automation
- *Dissertation:* **What is Common Knowledge Across Domains? Investigating Shared Representations in Transfer Learning** *(Advisor: Dr. Siddhartha Bhattacharyya)*

Florida Institute of Technology — *Melbourne, Florida*
M.S. IN COMPUTER SCIENCE: GPA - 4.0/4.0 — *May 2022 - December 2023*

- Core subjects: Artificial Intelligence, Database Systems, Formal Methods, Advanced Software Engineering, Speech Recognition.

Birla Institute of Technology and Science, Pilani — *Dubai, United Arab Emirates*
B.E. IN COMPUTER SCIENCE: GPA - 3.5/4.0 — *August 2018 - December 2021*

- Thesis: *ERF-CondLaneNet: an Ego Car Lane Detection Framework*

Skills

Programming	Java, C/C++, Python, Visual Basic, Bash, MATLAB, MySQL, MongoDB, LaTeX
Web & API Development	Django, FastAPI, Node.js, PHP/Apache, REST APIs
Frontend	HTML5, CSS3, JavaScript, React.js, JQuery
Data Analytics	Jupyter, pandas, Dask, Statsmodels, Seaborn, MySQL Workbench, Neo4j, ElasticSearch
Machine Learning Libraries	scikit-learn, XGBoost, OpenCV, nltk, pandas, scipy
Deep Learning / LLMs	TensorFlow, PyTorch, Keras, Cuda, TensorRT, BERT, GPT, Hugging Face Transformers, Hugging Face Accelerate, PEFT/LoRA, LangChain, LlamaIndex
Cloud & MLOps	AWS (S3, SageMaker, EC2), Ray, Weights & Biases, MLflow
Formal Verification	NuSmv/NuXmv, Uppaal, AGREE, PRISM, TLA+ - Coq (Class Projects)
Robotics & Systems Engineering	Robot Operating System (ROS), AADL, SysML, Gazebo, RViz, Agile Software Development
Other Tools	Docker, Git, Perforce, GitHub Actions, NLP Modules, SparkAR Studio, Android Studio, Arduino
Project Tools	Jira, Trello, Miro, Notion, Slack, Teams

Experience

Software and Systems Engineering, Avidyne — *Melbourne, FL*
SOFTWARE ENGINEER II — *May 2026 – Present*

- Design and implement **embedded C++ software** for avionics flight display systems, spanning **operating system services**, **device drivers**, and **touchscreen-based user interfaces**.
- Verify embedded software correctness through **unit and integration testing**, in accordance with **FAA** and internal certification standards.
- Maintain requirements-to-implementation traceability using **DOORS** to support certification audits.
- Enhance internal development tools and automation frameworks, improving efficiency for the broader engineering team.
- Mentor junior engineers and interns, guiding onboarding and technical growth.

Software and Systems Engineering, Avidyne — *Melbourne, FL*

SOFTWARE ENGINEER INTERN / CO-OP — *May 2025 – May 2026*

- Expanded **end-to-end (E2E) test and evaluation (E2TE)** coverage across navigation, communication, flight management, and flight display systems, increasing overall test coverage by **25%**.
- Developed, integrated, and debugged **C-based flight software** in the avionics stack, ensuring compliance with real-time, safety-critical **DO-178C** guidelines for production-ready systems.
- Built **30+ system-level and flight-specific test cases** and **TFC (Test Framework Core)** software tests, reducing initial verification cycle time by **10%**.
- Consolidated product test libraries across multiple avionics platforms into a unified cross-product testing infrastructure, improving code reusability and maintenance efficiency by **30%**.
- Automated regression and validation testing by writing **C and Visual Basic** test scripts (managed via **Perforce** and **Visual Studio**), cutting simulation turnaround time by up to **20%**.
- Diagnosed and resolved execution issues in **hardware-in-the-loop (HIL) simulations** using proprietary avionics simulation frameworks, improving simulation-to-aircraft fidelity by **10%**.
- Contributed to flight code development for Avidyne's **Quantum Open Avionics Platform**, supporting rapid prototyping of next-generation avionics solutions.

ASSIST Research Lab, Florida Institute of Technology — *Melbourne, FL*

RESEARCH SCHOLAR — *August 2021 - Present*

- Working with a team of research professionals for formal methods of verification and run-time assurance, ML, IoT, robotics, and cyber security.
- Contributed with **professors on the development of research proposals** on diverse topics, including cognitive agents on human behavior, the assurance of artificial intelligence in safety-critical systems, and the fine-tuning of LLMs for domain-specific queries.
- Collaborating on NASA's ***University Leadership Initiative (Round 8)*** as part of a **Florida Tech**-led multi-university/industry team to develop a framework for **trustworthy, increasingly autonomous aviation safety systems**; partners include ***Penn State, NC A&T State, UF, Stanford, Santa Fe College, Uni of New Mexico, Collins Aerospace, and ResilienX***; part of awards totaling up to $6.9M over three years.
- Collaborating with ***Collins Aerospace, Iowa State University, RTX Technologies Research Center (RTRC), and Smart Information Flow Technologies (SIFT)***, funded by ***DARPA*** with the task of formally modeling human cognitive behavior representation with respect to cyber-sickness in AR/VR systems.
- Collaborated with ***Penn State University*** on the application of **Large Language Model (LLM) translation** for cognitive architectures, focusing on enhancing the integration of LLMs to facilitate communication and knowledge transfer within cognitive systems.
- Contributed with ***Critical Frequency Design***, funded by ***Naval Air Systems Command***, with the task of developing a modeling approach for designing, maintaining, and supporting air and sea platform fiber optic communications technology.
- Collaborated with ***Rockwell Collins and Soar Tech***, funded by ***NASA*** with the task of formally verifying the autonomous agent to assure safety as well as the logical correctness of the safety-critical system.
- Advising a Ph.D. student and undergraduate students on transfer learning, automated data labeling, and assurance frameworks for vision-based classification in autonomous aircraft systems funded by ***NASA***, addressing safety and reliability challenges in aviation technologies.
- Mentoring **undergraduate** and **high school students** on machine learning engineering approaches in the aerospace and systems domains, leading to **conference publications** that addressed real-world challenges in these fields.
- Investigated the development of a **cognitive-enhanced agent** for automatically piloting aircraft in dense urban environments which emphasized safe and reliable takeoff/landing among aerial traffic without human intervention.
- Collected and presented **AssistTaxi**, a novel dataset for runway and taxiway analysis, contributing to autonomous operations.
- Developed an **autonomous aircraft perception system** for accurately detecting and labeling line markings on an airport taxiway.
- Recognized with multiple honors and leadership roles, including **Outstanding Student of the Year** at Honors Convocation 2025, **Inducted Member of Phi Kappa Phi**, and **President of the Florida Tech Badminton Club**.

L3Harris Institute for Assured Information, Florida Institute of Technology — *Melbourne, FL*

GRADUATE RESEARCH ASSISTANT — *May 2024 – June 2024*

- Developed a **decentralized framework** enabling **multiple autonomous agents**—robotic dogs, drones, and mobile robots—to coordinate, communicate, and reach shared goals.
- Collaborated with **developers and professors** to rigorously test the system in both **simulation** and **real-world environments**.

IRI Research, Florida Institute of Technology — *Melbourne, FL*

IRI RESEARCH DEVELOPER — *May 2023 - July 2023*

- Proposed and **implemented a framework using AI language models** to automatically extract software requirements from source code.
- Supervised and coordinated with undergraduate students towards the development process.

Publications

Modular Test-time Input-Space Refinement for Few-Shot Segmentation

MAH KHAN, **P GANERIWALA**, A ALVAREZ AND S BHATTACHARYYA

Under Review
IEEE Trans on Emerging Topics in Computational Intelligence

Surveying the Landscape of Transfer Learning: Common Knowledge and Beyond

P GANERIWALA, N NEOGI AND S BHATTACHARYYA

Under Review
IEEE Trans on Pattern Analysis and Machine Intelligence

Formalizing Natural Language Specifications for MBSE with an NLP-Based Grammar for Automated System Architecture Generation
Under Review
C CHAMBERS, **P GANERIWALA**, S MUELLER, S BHATTACHARYYA AND C SEN
IEEE Systems Journal

Formal Analysis of Stochastic Cognitive Models: A Translation Framework from Soar to PRISM
Accepted
P GANERIWALA, C CHAMBERS, JI LATHROP, S BHATTACHARYYA, J BABAR, S GILBERT, I AMUNDSON, M DORNEICH AND MAH KHAN
NASA Formal Methods 2026

Compositional Reasoning over System Architectures with Integrated Cognitive Model
Accepted
P GANERIWALA, C CHAMBERS, S BHATTACHARYYA, I AMUNDSON, AND J BABAR
IEEE SysCon 2026

Correlation-Threshold Feature Selection for Internet of Things (IoT) Device Identification under Domain Shift
Accepted
A DHANAWADE, **P GANERIWALA**, AND S BHATTACHARYYA
ISAIM 2026

AssistTaxi-v2: A Scalable Dataset for Taxiway/Runway Scene Understanding Under Diverse Conditions
Accepted
P GANERIWALA, MAH KHAN, A ALVAREZ, S BHATTACHARYYA, N NEOGI AND S LEHMAN
AIAA SciTech 2026

Explainable Assurance through Compositional Verification with Cognitive Models
Accepted
P GANERIWALA, C CHAMBERS, S BHATTACHARYYA, AND J BABAR
IEEE RTSS/ESRA 2025

Evaluating LLM Translation for Prompt-Enhanced ACT-R and Soar Models
Accepted
P GANERIWALA, S WU, S BHATTACHARYYA AND F RITTER
BRIMS 2025

Enabling Formal Verification in a Common Model of Cognition
Accepted
P GANERIWALA, M MATSUMURO, F RITTER AND S BHATTACHARYYA
BRIMS 2025

Integrating Reconfigurable Accelerators with Quantum Computing
Accepted
PRATIBHA, **P GANERIWALA** AND N MAHMUD
IEEE QCE QCore Workshop 2025

Adapt, But Don't Forget: Fine-Tuning and Contrastive Routing for Lane Detection under Distribution Shift
Accepted
MAH KHAN, **P GANERIWALA**, S LEHMAN, A ALVAREZ, S BHATTACHARYYA AND N NEOGI
ICCV 2COOOL Workshop 2025

AI Driven Differentiation and Quantification of Metal Ions Using ITIES Electrochemical Sensors
Accepted
M AHMED, **P GANERIWALA**, A SAVVIDOU, N BREEN, S BHATTACHARYYA, P PATHIRATHNA
Journal of Sensor and Actuator Networks 2025

FLAIR: Few-Shot Learning for Grapheme Recognition in Ancient Scripts
Accepted
P GANERIWALA AND D MITRA
CVPR SINT4CH Workshop 2025

Few-Shot Learning for Grapheme Recognition in Ancient Scripts
Accepted
P GANERIWALA AND D MITRA
ACM Journal on Computing and Cultural Heritage 2025

Can Someone Prove Your Operator Won't Get Distracted? A Gentle Introduction to Formal Methods in Human Factors
Accepted
S GILBERT, **P GANERIWALA**, J LATHROP, A NEWENDORP, S FIEFFER, P WU, I AMUNDSON, C CHAMBERS, A KOHL, S KHAN, M SANAEI, J BABAR, T WANG, D MUSLINER, R GOLDMAN, J GOTTLIEB, S GILBERT, E WINER, M DORNEICH AND S BHATTACHARYYA
HFES 2025

Modeling and Formal Analysis of High-Assurance Mixed-Reality Systems
Accepted
I AMUNDSON, J BABAR, H HERENCIA-ZAPANA, S F ROLLINI, B BRUSSEE, P WU, T WANG, D MUSLINER, R GOLDMAN, J GOTTLIEB, A NEWENDORP, A KOHL, S FIEFFER, S KHAN, M SANAEI, M MUSCALA, S GILBERT, E WINER, M DORNEICH, J LATHROP, **P GANERIWALA**, C CHAMBERS AND S BHATTACHARYYA
AIAA DATC/IEEE DASC 2025

Systems Engineering with Architecture Modeling, Formal Verification and Human Interactions for Learning-Enabled Autonomous Agent — *Accepted*
P GANERIWALA, R JONES, M MATESSA, S BHATTACHARYYA, J DAVIS, S ROLLINI, H PUROHIT, N NEOGI — *INCOSE Systems Journal*

Design and Validation of Learning Aware HMI for Learning-Enabled Increasingly Autonomous Systems — *Accepted*
P GANERIWALA, M MATESSA, S BHATTACHARYYA, R JONES, J DAVIS, P KAUR, S ROLLINI, N NEOGI — *IEEE SysCon 2025*

Automating Physics-Based Reasoning for SysML Model Validation — *Accepted*
C CHAMBERS, S MUELLER, **P GANERIWALA**, S BHATTACHARYYA AND C SEN — *IEEE SysCon 2025*

Runway vs. Taxiway: Challenges in Automated Line Identification and Notation Approaches — *Accepted*
P GANERIWALA, A ALVAREZ, A ALQAHTANI, S BHATTACHARYYA, MAH KHAN, N NEOGI — *IEEE SysCon 2025*

Exploring Machine Learning Engineering for Object Detection and Tracking by Unmanned Aerial Vehicle (UAV) — *Accepted*
A GUNA, **P GANERIWALA**, AND S BHATTACHARYYA — *IEEE ICMLA 2024*

ALINA: Automated Line Identification and Notation Algorithm — *Accepted*
MA H KHAN, **P GANERIWALA**, S BHATTACHARYYA, N NEOGI AND R MUTHALAGU — *CVPR VDU Workshop 2024*

AssistTaxi: A Comprehensive Dataset for Taxiway Analysis and Autonomous Ops — *Accepted*
P GANERIWALA, S BHATTACHARYYA, S GUNTHER, B KISH, MA H KHAN, A DHADOTI AND N NEOGI — *IEEE ICMLA 2023*

Towards Knowledge Extraction and Parsing of XML Metadata for SysML System Architecture Modeling — *Accepted*
C CHAMBERS, **P GANERIWALA**, S BHATTACHARYYA, C SEN AND N NUR — *IEEE UEMCON 2023*

Automated Framework to Extract Software Requirements from Source Code — *Accepted*
C MISKELL, R DIAZ, **P GANERIWALA**, K SLHOUB, F NEMBHARD — *ACM NLPIR 2023*

Assuring Learning-Enabled Increasingly Autonomous Systems (ALEIAS) — *Accepted*
N NARAYAN, **P GANERIWALA**, R JONES, M MATESSA, S BHATTACHARYYA, J DAVIS, H PUROHIT AND S ROLLINI — *IEEE SysCon 2023*

IPAssess: A Protocol-Based Fingerprinting Model for Device Identification in IoT — *Accepted*
P GANERIWALA, S NANDANWAR, A GUPTA, S BHATTACHARYYA AND R MUTHALAGU — *SAI IntelliSys 2023*

Cross Dataset Analysis and Network Architecture Repair for Autonomous Car Lane Detection — *Accepted*
P GANERIWALA, S BHATTACHARYYA AND R MUTHALAGU — *IEEE Intelligent Vehicles Symposium (IV) 2023*

Functional Reasoning of System Architecture in the System Modeling Language (SysML) With XML Representation — *Accepted*
P GANERIWALA, C CHAMBERS, C SEN AND S BHATTACHARYYA — *ASME IDETC-CIE 2023*

Modeling IoT Behavior for Enforcing Security and Privacy Policies — *Accepted*
A GUPTA, D CAMPOS, A DCOSTA, **P GANERIWALA**, S BHATTACHARYYA AND T OCONNOR — *SAI Computing Conference 2022*

Towards Generating System Arch and Formal Functional Description in AADL — *Accepted*
A CHAUHAN, **P GANERIWALA**, C SEN AND S BHATTACHARYYA — *ASME IDETC-CIE 2022*

Program Committees

Feb 2026	**Invited Review Committee Member,** International Conference on Artificial Intelligence, Computer, Data Sciences and Applications	*Boracay Island, Philippines*
Dec 2025	**Invited Reviewer,** 24th IEEE International Conference on Machine Learning and Applications (ICMLA)	*Boca Raton, FL*
Oct 2025	**Invited Reviewer,** 18th International Conference on Social Computing, Behavioral-Cultural Modeling & Prediction and Behavior Representation in Modeling and Simulation.	*Pittsburgh, PA*

Feb 2025	**Invited Reviewer: Top Reviewer**, NeurIPS 2025	*San Diego, CA*
Dec 2024	**Invited Reviewer**, 23rd IEEE International Conference on Machine Learning and Applications (ICMLA)	*Miami, FL*
Dec 2024	**Invited Reviewer**, International Conference on Electrical and Computer Engineering Research	*Gaborone, Botswana*
Dec 2024	**Invited Reviewer**, International Conference on Machine Learning (ICML) 2025	*Vancouver, Canada*
Nov 2024	**Invited Reviewer**, The 4th International Conference on Electrical, Computer, Communications and Mechatronics Engineering	*Male, Maldives*
Sep 2024	**Invited Reviewer**, International Conference on Learning Representations (ICLR) 2025	*Singapore EXPO*
Aug 2024	**Invited Reviewer**, 23rd IEEE International Conference on Machine Learning and Applications	*Miami, FL*
Jun 2024	**Invited Reviewer**, International Conference on AI, Information Processing and Cloud Computing	*Sanya, China*
May 2024	**Invited Reviewer**, NeurIPS 2024	*Vancouver, Canada*
Aug 2023	**Invited Reviewer**, 22nd IEEE International Conference on Machine Learning and Applications (ICMLA)	*Jacksonville, FL*
Nov 2022	**Invited Reviewer**, 2023 American Control Conference (ACC)	*San Diego, CA*